Reference

This article offers a comprehensive reference on brightness data and communication protocols, focusing on the Modbus-RTU protocol used in sensor operations, highlighting key parameters, data frame formats, and register functions.

Brightness data reference:

At night: 0.001-0.02 lx;

Moonlit night: 0.02-0.3 lx;

Cloudy indoor: 5-50 lx;

Cloudy outdoor: 50-500 lx;

Sunny indoor: 100-1000 lx;

Summer noon light: about 10^6 lx;

Illumination when reading books: 50-60 lx;

Standard illumination for home video: 1400 lx;

Communication Protocol Description

1. Basic communication parameters

Interface	Encoding	Data bits	Parity bits	Stop bits	Error checking	Baud rate
RS485	8-bit binary	8	None	1	CRC	1200、2400、4800、9600、19200、38400、57600 bit/s configurable, default 9600bit/s

2. Data frame format definition

Using Modbus-RTU communication protocol, the format is as follows:

Initial structure ≥4 bytes of time

Address code = 1 byte

Function code = 1 byte

Data area = N bytes

Error check = 16-bit CRC code

End structure ≥4 bytes of time

Address code: The address of the sensor, which is unique in the communication network (factory default 0x01).

Function code: The function indication of the command sent by the host. This sensor reads the register function code 0x03 and writes the register function code 0x06

Data area: The data area is the specific communication data. Note that the high byte of 16-bit data is in front!

CRC code: A two-byte check code.

Register address

Register address	Content	Operation	Range and definition
0002H	High 16-bit data of illumination value	Read-only	Two bytes
0003H	Low 16-bit data of illumination value	Read-only	Two bytes
0046H	Illumination acquisition rate	Read-write	Level 1-20
0047H	Illumination calibration enable bit	Read-write	01 Enable illumination calibration function, 00 Disable illumination calibration function
0048H	Illumination calibration compensation value	Read-write	This value is written with the actual value enlarged by 100 times; for example, if compensation of 1.4 is required, the written value needs to be enlarged by 100 times, that is, 140; Calculation method: (collected illumination value ÷ ambient illumination value) × 100 = compensation value For example: the collected illumination value is 221, the actual ambient illumination is 260, and the calculated compensation value = (221 ÷ 260) × 100 = 85
0064H	16-bit device address	Read and write	Range is (1-254), 255 is the general control command address
0065H	16-bit baud rate selection	Read and write	0-1200, 1-2400, 2-4800, 3-9600, 4-19200, 5-38400, 6-57600
0066H	Parity bit	Read and write	00 no parity, 01 odd parity, 02 even parity
0067H	Version information	Read only	Level 1-20
00E0H	Device soft reset write command	Write only	Level 1-20
00F0H	Device factory reset write command	Write only	Level 1-20

3. Communication protocol example and explanation

3.1、Read multiple register illumination data instruction

Inquiry frame:

Address code	Function code	Register start address	Register length	Check code low bit	Check code high bit
0x01	0x03	0x00 0x02	0x00 0x02	0x65	0xCB

Response frame:

Address code	Function code	Return valid bytes	Illumination data high 16 bits	Illumination data low 16 bits	Check code low bit	Check code high bit
0x01	0x03	0x04	0x00 0x0C	0xF4 0x73	0x3D	0x15

Light value:

00 0C F4 73 (hexadecimal) =849011/1000 =>Light value = 849.011Lux

3.2, Modify the current address

Inquiry frame: (Modify the current address to 0x02)

Address code	Function code	Register address	Modify value	Check code low bit	Check code high bit
0x01	0x06	0x00 0x64	0x00 0x02	0x49	0xD4

Response frame:

Address code	Function code	Register address	Modify value	Checksum low bit	Checksum high bit
0x01	0x06	0x00 0x64	0x00 0x02	0x49	0xD4

3.3, Modify address, baud rate

Inquiry frame: (Modify address to 0x01, baud rate to 9600)

After modifying the address or baud rate, the sensor needs to be powered off and reconnected

Address code	**Function code**	**Register start address**	**Register length**	**Modify address**	**Modify baud rate**	**Parity check**	**Chec low b**
0x01	0x10	0x00 0x64	0x00 0x03	0x00 0x01	0x00 0x03	0x00 0x00	0xA9

Response frame:

Address code	**Function code**	**Register start address**	**Register length**	**Modify address**	**Modify baud rate**	**Checksum low bit**
0x01	0x10	0x00 0x64	0x00 0x03	0x00 0x01	0x00 0x03	0x9C

3.4, query current address, baud rate

Inquiry frame:

Address code	**Function code**	**Register start address**	**Data length**	**Check code low bit**	**Check code high bit**
0xFF	0x03	0x00 0x64	0x00 0x02	0x90	0x0A

Response frame:

Address code	**Function code**	**Return valid bytes**	**Device address**	**Baud rate**	**Check code low bit**	**Check code high bit**
0x01	0x03	0x04	0x00 0x01	0x00 0x03	0xEB	0xF2

The real address of the device read is 01, and the baud rate is 0x03, which is 9600.

3.5, Device factory reset command

Inquiry frame:

Address code	**Function code**	**Register address**	**Modify data**	**Check code low**	**Check code high**
0xFF	0x06	0x00 0xF0	0x00 0x00	0x9C	0x27

Response frame:

Address code	**Function code**	**Register address**	**Modify data**	**Check code low**	**Check code high**
0xFF	0x06	0x00 0xF0	0x00 0x00	0x9C	0x27